MAYO CLINIC PROCEEDINGS

THEMATIC REVIEW ON FORWARD THINKING ON CLINICAL TRIALS IN CLINICAL PRACTICE

Health Disparities, Clinical Trials, and the Digital Divide

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Abstract

In the past few years, there have been rapid advances in technology and the use of digital tools in health care and clinical research. Although these innovations have immense potential to improve health care delivery and outcomes, there are genuine concerns related to inadvertent widening of the digital gap consequentially exacerbating health disparities. As such, it is important that we critically evaluate the impact of expansive digital transformation in medicine and clinical research on health equity. For digital solutions to truly improve the landscape of health care and clinical trial participation for all persons in an equitable way, targeted interventions to address historic injustices, structural racism, and social and digital determinants of health are essential. The urgent need to focus on interventions to promote health equity was made abundantly clear with the coronavirus disease 2019 pandemic, which magnified long-standing social and racial health disparities. Novel digital technologies present a unique opportunity to embed equity ideals into the ecosystem of health care and clinical research. In this review, we examine racial and ethnic diversity in clinical trials, historic instances of unethical research practices in biomedical research and its impact on clinical trial participation, and the digital divide in health care and clinical research, and we propose suggestions to achieve digital health equity in clinical trials. We also highlight key digital health opportunities in cardiovascular medicine and dermatology as exemplars, and we offer future directions for development and adoption of patient-centric interventions aimed at narrowing the digital divide and mitigating health inequities.

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The advances in digital health care technologies in the past few years have revealed huge potential to revolutionize the practice of medicine. We have also seen a rapid expansion of digital tools for health care delivery and research, many of which were accelerated by the coronavirus disease 2019 (COVID-19) pandemic, including an exponential growth in telehealth services and pragmatic clinical trials powered by novel technologies to allow for remote study participation.[1-3] Although digital technologies have the potential to transform clinical research and health care as we know it, there is mounting concern about their potential to propagate health disparities on an even larger scale.

The National Institutes of Health (NIH) defines health disparities as differences in health outcomes negatively affecting disadvantaged populations.[4] These disadvantaged populations include "racial/ethnic minorities, socioeconomically disadvantaged populations, underserved rural populations, and sexual and gender minorities."[4] There is abundant evidence that therapies and interventions to improve health are less likely to trickle down to or lag behind in health disparity populations.[5] Despite the NIH Revitalization of Act of 1993 establishing guidelines for the inclusion of women and racial and ethnic minority groups into clinical research, these groups remain underrepresented in clinical trials.[6-11] Racial and

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ethnic diversity in clinical trials remains abysmally low, with the percentage of all minorities in published randomized controlled trials estimated at approximately 4% in 2019.[12] Nonetheless, there have been notable clinical trials in the field of hypertension with laudable representation of Black participants. The Hypertension Detection and Follow-up Program (HDFP) in 1979 (44% Black adults) revealed the clinical benefits of intensive antihypertensive therapy among Black men and women with hypertension.[13] The Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT) completed in 2002 (36% Black adults) revealed the effectiveness and clinical benefits of thiazide diuretics,[14] and the African American Study of Kidney Disease and Hypertension (AASK) which completed participant follow up in 2001 (100% Black adults)[15,16] was the first study to document the renoprotective effects of angiotensin-converting enzyme inhibitors in Black patients with kidney disease. There is also evidence that clinical studies conducted in health disparity populations have led to the discovery of novel therapeutics, for example, the landmark African-American Heart Failure Trial (A-HeFT), which revealed the effectiveness of hydralazine—isosorbide dinitrate in systolic heart failure,[17,18] or community-based interventions as seen with the hypertension control program in barbershops[19] and use of a culturally tailored mobile health (mHealth) lifestyle app to promote ideal cardiovascular health (Fostering African-American Improvement in Total Health! [FAITH!] Trial).[20] Although we point out these unique examples of landmark studies, adequate inclusion of racial and ethnic minority populations in clinical trials is still not the norm, and little progress has been made since the NIH mandate.[12]

In this review, we focus on the interrelationship between racial and ethnic health disparities (with particular focus on Black individuals), clinical trials, and the potential impact of digital technologies on widening or narrowing the disparity gap (Figure 1).

RACE AND ETHNICITY AS A VARIABLE IN CLINICAL RESEARCH TRIALS

Racism in Medical Research

Medical research is plagued by historic and pervasive racism, exploitation, experimentation, and brutal mistreatment of racial and ethnic minority populations. A few of the historic examples[21] of racial injustices in research include the United States Public Health Service—led Tuskegee Syphilis Study, the experimentation on enslaved Black women by Dr J. Marion Sims, the inappropriate use and distribution of the immortal cell line of Henrietta Lacks, and more recently, the misuse of DNA samples obtained from a Native American tribe.[22] Regrettably, institutionalized racism remains pervasive even in current times.[21] Clinical research and medical education also have a long and cruel history of racial exploitation and unethical practices[23] in the quest for more knowledge, such as the unofficial use of slave plantations as experimentation laboratories and the utilization of dead bodies of enslaved Black people in US medical schools for anatomic studies.[24,25] Scientific advancement and medical research have taken advantage of and benefited from racialization. These historic injustices set a harmful precedent and laid foundations for inequity in health care and participation in clinical trials.

Inappropriate Use of Race in Clinical Research

Race and racial identity are social constructs that emerged from prejudiced human perceptions, with no scientific basis, and when broadly applied in clinical research, the terms are vague and do not sufficiently account for innate biological differences or socioenvironmental influences.[26,27] However, the erroneous belief that race reflects biological differences has been used as a justification for colonization, slavery, and unethical research and, regrettably, continues to be propagated in biomedical research and literature. The underlying premise of racial injustice in health care reflects prevailing and dehumanizing myths

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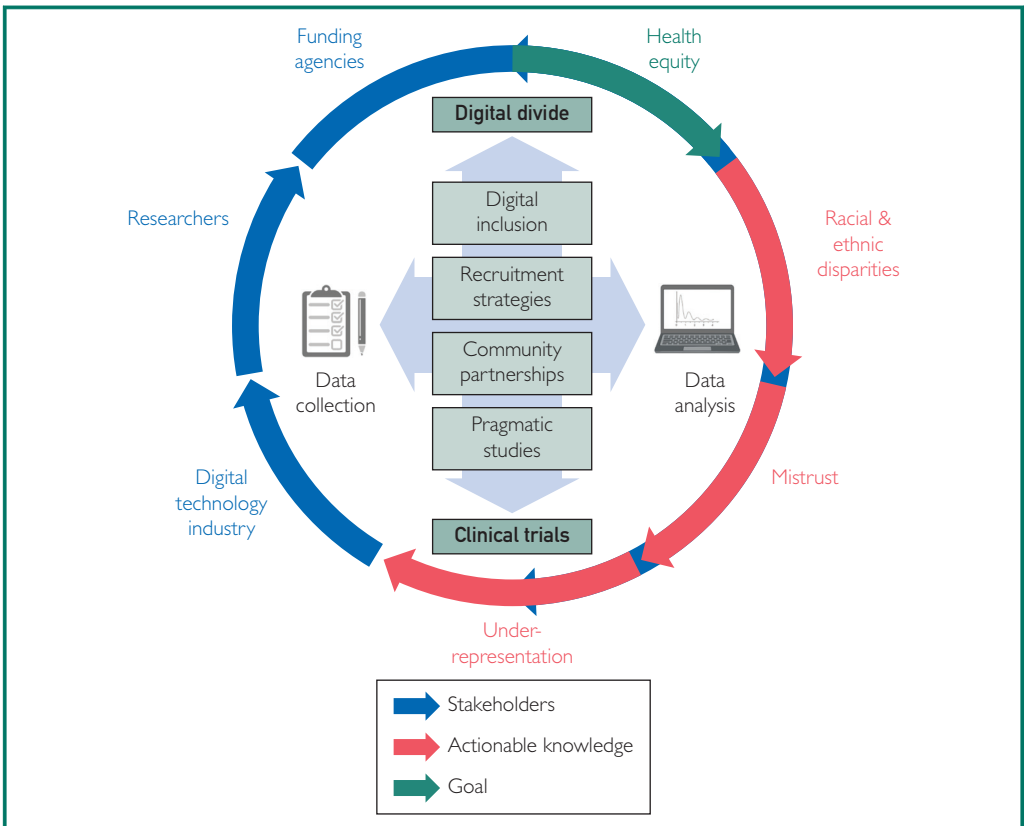


FIGURE 1. Health disparities, clinical trials, and the digital divide. Graph illustrates an interplay of factors influencing digital inequity, key stakeholders, and interventions to promote health equity in digital clinical trials.

about biological differences between Black and White persons, such as the assumption that Black patients are less likely to feel pain than White patients to justify disparate care. Notably, one study found that approximately half of White medical trainees surveyed endorsed false beliefs that Black patients are biologically different than White patients in relation to sensations of pain and skin thickness.[28]

Inappropriate use of race in medical research has also contributed to disparities in care.[29] For example, assessment of kidney function guides the safe management of renally excreted pharmacotherapeutics, but prespecified function thresholds also dictate inclusion/exclusion in clinical trials and eligibility for kidney transplant.[30] Estimated glomerular filtration rate, a measure of kidney function often calculated using the Chronic Kidney Disease Epidemiology Collaboration equation, includes a race multiplier term that results in a higher estimated glomerular filtration rate for Black patients compared with White patients with otherwise similar characteristics, which has negatively impacted early access to therapies such as kidney transplant.[31,32] Similarly race-based beliefs about variation in bone density by race can lead to inequities in osteoporosis screening and diagnosis.[33] Failure to personalize treatment decisions while relying on broad assumptions about race-related ineffectiveness of specific antihypertensive therapy types may have contributed to poorer blood pressure control among Black patients compared with other racial groups, whereas studies suggest there may be more variation within races than between races regarding antihypertensive therapy

response.[34] To advance equity, researchers must interrogate how racism (not race) contributes to disparate health outcomes.

Medical Mistrust

Knowing the history of racial injustice and exploitation of Black persons[35] in biomedical research and clinical trials informs a better understanding of the origins of medical mistrust, which has contributed to inadequate representation of racial and ethnic minority groups in clinical trials (Figure 1). Mistrust of the health care system has been reported to be a key barrier to participation in clinical research among Black patients and indigenous and rural populations.[36-39] However, other barriers have also been identified including health care costs, health literacy, digital literacy, limited access, and implicit bias against racial and ethnic minority groups.[6] Representation is extremely important in clinical trials because it provides essential data and scientific evidence to generalize the results of research studies and health care interventions to underserved and racial and ethnic minority groups. Unfortunately, the majority of clinical trials continue to fail in this regard even though these groups have a disproportionately higher burden of disease.[40]

It remains unknown whether the shift toward decentralized and pragmatic trials enabled by digital technology would be a stepping stone to addressing this issue. Strategies to increase enrollment of racial and ethnic minority groups in clinical trials research need to utilize a multipronged approach to ensure that patients are not exploited, must be socially and culturally responsive, should ensure adequate representation among clinical trial leadership, and include appropriate safeguards to move toward building trust in these communities.

THE DIGITAL LANDSCAPE IN HEALTH CARE AND CLINICAL RESEARCH

The Digital Divide

The *digital divide* is a term used to describe the widening inequalities between disadvantaged persons who do not have access to digital technologies (including computers, smartphones) or the internet and the more privileged individuals who do.[41,42] Digital redlining has also been identified as a barrier to health innovation. This novel term is defined as discriminatory policies and practices by communication technology providers in maintaining or upgrading infrastructure for service delivery in communities with racial and ethnic minority and socioeconomically disadvantaged groups.[43,44] The increased use of digital tools in clinical trials, especially as result of the COVID-19 pandemic, has prompted important questions about digital equity.[45] The use of digital tools can enhance the inclusion of underrepresented groups in clinical trials by decreasing recruitment and travel cost, facilitating remote data capture, and allowing patients to participate in studies with less impact on their day-to-day lives. However, digital disparities portend a new unique challenge to clinical trial participation. These disparities were made evident at the peak of the COVID-19 pandemic, which resulted in limited access to and lower utilization of telemedicine visits by patients from underserved communities.[46,47] In a large study of over 1500 primary and specialty care practices conducted between April and June 2020, older patients, racial and ethnic minority groups, and patients with limited English proficiency were less likely to use video visits.[48] Although technology can serve as a tool for advancing clinical trial equity, digital inequities need to be addressed first before this goal can be reasonably achieved.

Digital Equity and Inclusion

Digital equity is when "all individuals and communities have the information technology capacity needed for full participation in our society, democracy, and economy."[49] Digital equity and inclusion span several domains often referred to as *digital determinants of health*, and these domains include internet connectivity, digital technology access/affordability, and digital literacy that are now being recognized as key social determinants of health.[50] The digital health equity framework has been proposed as an avenue

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to incorporate a multilevel approach to address the digital determinants of health through acknowledgement of the digital environment domain (Figure 2). To effectively participate in digitally supported clinical trials, study participants need to be familiar with digital devices, have access to broadband internet, and be able to afford the associated cost of these technology-supported devices and services. Based on a 2019 report, approximately 21 million people in the United States lack broadband internet access.[51] Although mobile technology access in the United States has steadily increased over the years with 85% and 77% of adults in 2021 reporting they own a smartphone and home broadband subscription, respectively, up to 30% report having internet connectivity problems.[52] Among those without broadband internet access, 45% report a high monthly cost of home broadband subscription as a barrier to internet access. In addition, White adults were more likely to have home broadband than Black and Hispanic adults.[52] Recent policy initiatives, like the Infrastructure Investment and Jobs Act, provide funding for extending broadband services to underserved communities as well as internet and device subsidies.[44] These new resources could potentially address existing barriers to digital clinical trial participation. Access to digital technology in addition to digital literacy (defined as having the cognitive and technical skills to effectively utilize information and communication technologies[5,49]) can promote successful participation in clinical trials. Training of digital health navigators to assist patients with using digital tools and their integration into clinical practice has been proposed.[53] Digital literacy training, with or without the use of navigators, could potentially be adapted and included as part of the clinical trial recruitment and enrollment process and can in turn enhance digital self-efficacy and confidence. Further, equity in digital literacy relies not only on study participant factors but also on the usability and design of technology platforms. A study evaluating the usability of patient-facing mHealth apps found that patients or caregivers were able to complete only 43% of assigned tasks across 11 different apps without assistance.[54] This problem was, in part, due to poor app usability and design. This finding highlights an important opportunity for clinical trialists, researchers, and technology developers to collaborate in inclusive and human-centered design approaches for digital tools integrated into clinical research.

To achieve digital health equity in clinical trials, additional steps should be in place to ensure procedures for recruitment, implementation, and deployment of digital interventions are carefully selected and managed to minimize potential bias. Clinical trialists should ensure that recruitment strategies relying on digital tools do not inadvertently exclude study participants who are digitally disadvantaged. For example, online patient portals are increasingly used as recruitment tools, but underserved patients are less likely to use patient portals.[55,56] By providing multimodal approaches to recruitment, clinical trialists can ensure inclusivity. Research teams should consider digital tools as central to equitable recruitment because evidence from studies in clinical practice settings suggest that health care professional bias may contribute to disparities in care offered through digital technologies.[57] In addition, leveraging community partnerships and engagement in addition to deliberate and intentional efforts to mitigate bias are essential steps toward ensuring digital health equity.[58]

Specialty Focus: Disparities in Cardiovascular Care

Cardiovascular disease remains the leading cause of death among men and women in the United States, despite an ongoing pandemic.[59] Racial and ethnic minority groups bear a disproportionate burden of cardiovascular disease, with Black persons having the highest cardiovascular mortality rate in the United States.[60,61] Reduced access to cardiovascular care has been documented in multiple studies among racial and ethnic minority groups, particularly among Black individuals.[62,63] The use of digital health technologies has shown some promise in

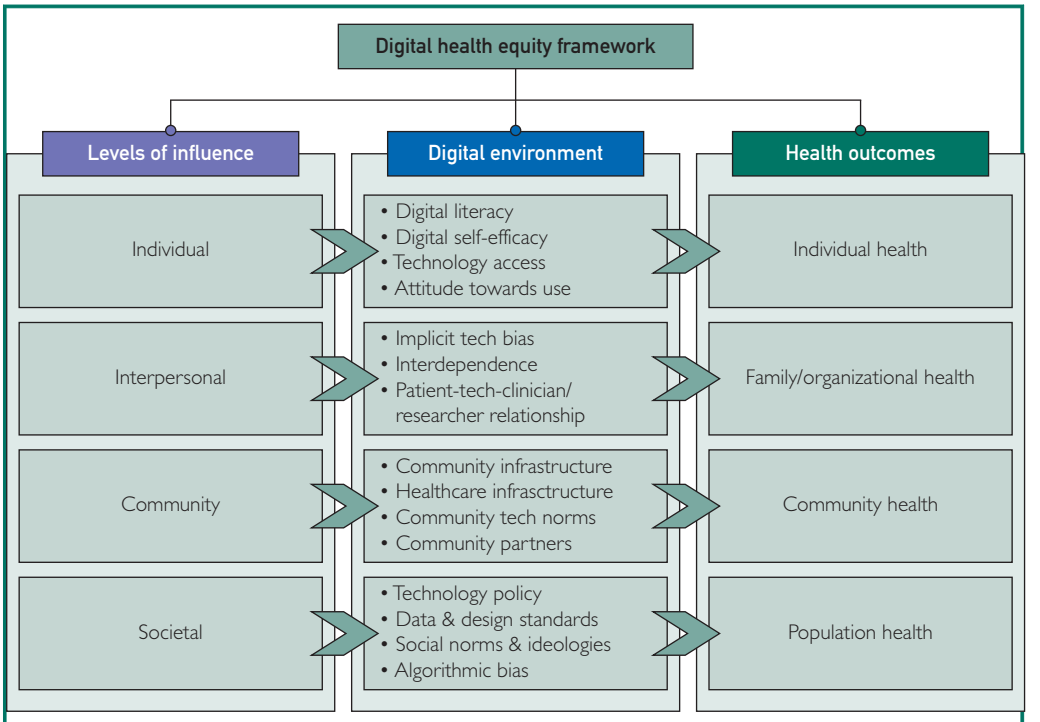


FIGURE 2. Proposed digital health equity framework based on the NIMHD Minority Health and Health Disparities Research Framework. Tech, technology/technician. Adapted from Richardson et al, *NPJ Digital Medicine*.[5] Creative Commons Attribution 4.0 International License.

improving cardiovascular health equity in certain populations, as reported in a large health system study of ambulatory cardiovascular clinics during the COVID-19 pandemic.[64] In this study, Black race was independently associated with higher telemedicine completion rates (with 11% higher odds of telemedicine visit completion).[64] Likewise, patients of low socioeconomic status, with median household income of less than $50,000, also had 22% higher odds of telemedicine visit completion in the cardiovascular clinics.[64] Similar findings were revealed in another study evaluating primary care appointments with lower no-show rates and higher visit completion rates for telemedicine visits during the COVID-19 pandemic as compared with in-person visits prior to the pandemic among Black patients.[65] Digital health technologies, therefore, represent a potential opportunity to improve access to health care and inclusion in cardiovascular clinical trials among marginalized patient groups that have traditionally faced barriers to accessing in-person care. When designing and refining digital health implementation in cardiovascular clinical trials, interventions should be tailored to and evaluated by those who are the most marginalized by our health care system. One study found that accessing online patient portals or smartphone applications may be prohibitive for patients from marginalized groups, whereas text-based approaches for cardiovascular disease management had higher uptake and were a more effective way for patients to engage with health care professionals.[66] The use of a community-based mHealth app has also shown benefits in improving ideal cardiovascular health among Black individuals.[20]

There are also notable inequities in access to and completion of cardiac rehabilitation (CR) among adults with cardiovascular disease.[67] Referral for CR is a class I recommendation by the American Heart

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Association and the American College of Cardiology for patients undergoing heart surgery or coronary intervention, those with stable angina, peripheral artery disease, or heart failure, and for myocardial infarction survivors prior to hospital discharge.[68,69] However, in a study of Medicare patients across the United States, racial and ethnic minority groups were less likely to receive CR referral compared with White patients.[70] Women are also less likely than men to be referred for, enrolled in, or complete CR.[71] Awareness of these inequities subsequently led to innovative strategies to increase CR enrollment and participation that utilize digital technologies for the delivery of home-based[69] and virtual[72,73] CR. In addition, the use of gamification and food-tracking apps to enhance CR[74] have shown promising results.

The use of machine learning (ML) and artificial intelligence (AI) in cardiovascular research has also had impressive diagnostic performance,[75] and some models have outperformed the current standard of care.[76] As we move toward potential implementation of these ML models, it is important that training data sets are inclusive and thoroughly validated in diverse patient cohorts through prospective studies and clinical trials to guard against algorithmic bias. Although some studies have made effort in this regard,[77,78] evidence from clinical trials remains limited. An ongoing clinical trial (ClinicalTrials.gov Identifier: NCT05438576) aims to evaluate the potential role for AI-powered technologies and portable digital electrocardiography tools to screen for pregnancy-related cardiomyopathies with the goal of narrowing the disparity gap in maternal cardiovascular mortality. This AI tool will be evaluated in a predominantly Black population of women in Nigeria who have the highest reported incidence of peripartum cardiomyopathy worldwide.[79] If proven to be effective, the results of this study will be a game changer for the field of cardio-obstetrics in the United States and internationally. The lessons learned from successful studies utilizing digital tools for CR and clinical research, particularly among women and racial and ethnic minority groups, may provide useful insights in planning equitable and pragmatic clinical trials using digital technologies.

Specialty Focus: Disparities in Dermatologic Care

An estimated 3 billion people in the world have inadequate access to dermatologic care.[80] In the United States, dermatologists are unequally distributed geographically, with counties with a high density of minorities having more shortages of dermatologists.[81] Limited access to dermatologic care has negatively impacted patient outcomes, including mortality, for racial and ethnic minority groups and people of low socioeconomic status for various dermatologic diseases such as malignant melanoma and Merkel cell carcinoma.[82] Leveraging digital innovations to help increase access, improve diagnostic accuracy, and improve efficiency of dermatologic care has been proposed as a solution to these challenges. However, if not developed with inclusivity in mind, these solutions may present additional roadblocks that could exacerbate health care disparities in dermatology.

Dermatology is a visually oriented specialty that greatly relies on pattern recognition, making it a great candidate for ML-driven innovation. Because of the large clinical, dermatoscopic, and histopathologic image database available in dermatology, there has been a rapid increase in the number of AI studies focused on early diagnosis of a wide variety of skin disorders. Integration of AI-based tools with digital technologies such as smartphone cameras and apps has opened a new era of access to dermatologic care.

The ability to successfully provide reproducible, nonbiased results with ML depends on high-volume and high-quality data. In the case of dermatology, this means training algorithms to learn the manifestation of skin diseases in all skin types. Currently, most images used to train and test algorithms are lacking in diverse skin tones.[83] For instance, the International Skin Imaging Collaboration data set, which is one of the

largest and most often used open-source data sets, lacks clinical images of diverse skin tones.[83,84] This issue highlights the concern that ML programs are largely learning on lighter skin tones. In a recent study of an AI-based tool developed for the diagnosis of dermatologic conditions, it was noted that patients with Fitzpatrick type VI (darkest skin type) skin were absent from the data used to develop the algorithm.[85] Therefore, no matter how advanced the ML algorithm, it may underperform when presented with images from patients with darker skin.[83]

To develop equitable, generalizable AI in dermatology, data sets and clinical trials that include a diversity of skin tones and a range of diseases are needed. Image repositories around the world need to include photos of skin conditions in darker skin types. The Diverse Dermatology Images (DDI) data set was created to provide a publicly accessible data set with diverse skin tones to be used in AI development and testing to ensure that algorithms are being appropriately trained to address the dermatologic needs of racial and ethnic minority patients.[86] Without the intentionality of AI training and testing using diverse data, this new technology solution has the potential to leave marginalized groups behind and further exacerbate existing disparities in the dermatologic care of patients with darker skin[83] Specific guidelines for dermatology AI development can help improve data transparency and provide a mechanism to assess bias in AI applications for dermatology.

The adoption of teledermatology was born from a need to tackle the challenges of inadequate access to dermatologic care. However, its rapid expansion has been fueled by the COVID-19 pandemic.[87] Although the geographic disparities in access to dermatology contribute to the increased use of teledermatology, this expansion has also made evident the importance of digital literacy for empowering patients to engage productively with this new care model.[44] Specifically, patients who are older, non—English-speaking, or of lower socioeconomic status face additional difficulties with access to teledermatology, particularly video visits.[44] Equitable care delivery and access to digital clinical trials among underserved patients require an inclusive system that ensures access to affordable broadband, internet-enabled devices, digital-literacy support, and appropriately designed platforms.[88] In addition, the efficacy of teledermatology relies on dermatologists with knowledge of diverse skin types and cultural competence. This issue is particularly important because skin disease in patients from racial and ethnic minority groups can appear different from manifestations in White patients and may not be easily diagnosed. Therefore, the need to prioritize undergraduate and postgraduate medical education on recognizing skin diseases on diverse skin types as well as prioritizing inclusion of diverse skin types in clinical research and trials has never been more urgent. As implementation of digital innovations and technologies to expand dermatologic care continues to advance, ensuring equitable delivery among diverse communities remains of paramount importance.

FUTURE DIRECTIONS

Community Engagement/User-Centered Design of Digital Innovations for Clinical Trials to Promote Health Equity

To address the escalating urgency of health disparities and promote health equity, more inclusive strategies through community engagement as well as participatory and user-centered design of digital innovations alongside socioeconomically disadvantaged and underresourced communities are essential.[58] The integration of community preferences within digital interventions to enhance their sociocultural relevance has been found to also improve their overall utility, acceptability, and adoption among specific communities, particularly those that have been historically marginalized in clinical research.[89] These sociotechnical interventions can have a major impact on health disparities because they place value on the diversity of patient experiences to improve outcomes through multilevel

approaches.[90,91] Community partners are at the heart of these approaches because they play a major role in shaping the future digital landscape of their communities and increasing access to clinical trials. Authentic partnerships with advocacy groups (focused on technology, public health, medical access) and trusted community leaders and messengers (community health workers) are key to achieving digital inclusion. In turn, these partnerships can mitigate mistrust of researchers and empower patients to participate in research that ultimately benefits their communities and society.

Role of Federal Agencies, Industry, and Academia

Key stakeholders in bridging the digital divide include federal agencies, industry, and academic institutions because they can directly impact the digital determinants of health at the societal level[5] (Figure 1) by addressing structural racism and systemic oppression. They shape policies for broadband internet access, the technological health care infrastructure, and the topography of clinical research and clinical trial funding arenas that can either exacerbate or reduce existing health disparities. The expansion and sustainability of federal programs and policies to support more affordable and high-quality availability of broadband connectivity, telecommunications, and digital health tools (eg, mobile devices, remote patient monitoring) are crucial to combating digital redlining of impoverished neighborhoods.[43] Further, policymakers must place more attention on improving high-quality health care delivery, especially remote care through telehealth within marginalized communities. Otherwise, these populations will be left behind with limited digital access and adverse health outcomes. In addition, as the digital ecosystem grows, so will clinical research and clinical trial networks. Federal agencies must prioritize funding for rigorously designed research grounded in implementation science and health equity principles along the clinical and translational pipeline.[92] These studies should not only generate evidence to support safety and efficacy of digital and nondigital interventions but also normalize the dissemination of study results and effective real-world solutions to participants and their communities. Researchers in industry and academia must be intentional in including marginalized and health disparity populations and ensure equal access through innovative trial designs (eg, stepped-wedge and delayed intervention randomized controlled trials, community-oriented recruitment/implementation).[19,20,93,94] Further, it is crucial that funding agencies, regulators, and academic leadership operationalize and monitor health equity by incentivizing investigators actively working toward achieving diversity in clinical trials.[95]

Leveraging Digital Tools to Improve Clinical Trial Diversity

There is promise in the incorporation of digital tools to enhance clinical trial diversity and inclusion of health disparity populations. However, it will take concerted and strategic efforts to simultaneously change individual adoption of digital tools, particularly among underserved populations.[93] This change includes enhancing their acceptance of and willingness to use digital technologies while also building their trust in research participation. In the same manner that digitization of health care may democratize the "patient-tech-clinician" relationship to a more equalized partnership through data transparency and access,[5] digitization of clinical research could transform the "patient-tech-researcher" relationship (Figure 2) by fostering trust, resulting in more meaningful research and equitable opportunities for participation in clinical trials. Health care ecosystems can equip and enable broader patient inclusion in research via digital applications and approaches that are more inclusive to diverse groups (eg, electronic consenting platforms, minimizing in-person interactions, home health care visits, and at-home self-collection kits, digital biomarkers).[89] Lastly, the value of digital health and clinical research should

hinge upon more than the simple question, "Does the solution work?"[96] but also "Who is represented, and why?"[97] and "In whom can we make the most difference?" This realignment on social justice is the only way that we will truly achieve digital health equity and diversity in clinical trials.

CONCLUSION

The use of digital technologies has the potential to transform clinical trials and at the same time address health disparities if implemented responsibly. As these technologies expand and become more mainstream in the clinical trial landscape, it is essential that appropriate safeguards are put in place to ensure that disparities are not inadvertently reinforced and that trial efforts strive toward the goal of achieving health equity.

POTENTIAL COMPETING INTERESTS

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SUPPLEMENTAL ONLINE MATERIAL

Supplemental material can be found online at http://www.mayoclinicproceedings.org. Supplemental material attached to journal articles has not been edited, and the authors take responsibility for the accuracy of all data.

Abbreviations and Acronyms: **AI,** artificial intelligence; **COVID-19,** coronavirus disease 2019; **CR,** cardiac rehabilitation; **mHealth,** mobile health; **ML,** machine learning; **NIH,** National Institutes of Health

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